

CLOSING THE ALGEBRAIC LOCUS OF WEIL CLASSES: AN ATTEMPT AND THE EXACT POINT WHERE IT STOPS

DEEP BHATTACHARJEE

ABSTRACT. On a general abelian variety of Weil type the Hodge conjecture comes down to one question: is the Weil class algebraic? The set of members of a Weil family on which it is algebraic is known to be dense, and to be either the whole family or a meagre set. This note tries to show that it is the whole family, by the most direct route: take a member where the class is algebraic and move the cycle to every other member. We prove what that route gives. A stratum of the algebraic locus that passes through a member with the largest possible Hodge group has the largest possible monodromy. A single semiregular cycle at any one member would close the locus. We then show where the route stops. For most fields and discriminants the only members at which the class is known to be algebraic have small Hodge group. For $\mathbb{Q}(\sqrt{-3})$ and a split form, Schoen's cycles on Prym varieties reach members with full Hodge group, but only along a locus of dimension $3n$, which is smaller than the family once $n \geq 4$. No semiregular cycle with the right class is known. The attempt does *not* prove the Hodge conjecture for these varieties, and we say precisely what is missing. We then treat the problem as an instance of the variational Hodge conjecture and prove three further results that narrow where a proof can come from. The first two are variants of arguments in the author's earlier work, and the third is new: the locus is everything exactly when the class has representatives of bounded degree on a Zariski-dense set; at a member with full Hodge group no class or Chern character built from divisors and homomorphisms can reach the Weil class; and the same holds for the tautological cycles of curves with an automorphism of order three.

1. WHAT WE ARE TRYING TO PROVE

Fix an imaginary quadratic field K and an integer $n \geq 2$. An abelian variety A of dimension $2n$ is of *Weil type* for K if K acts on it by endomorphisms (up to isogeny), the Rosati involution of a polarisation η acts on K as complex conjugation, and the two eigenspaces of K on the tangent space of A both have dimension n . Then $V = H^1(A, \mathbb{Q})$ is a K -vector space of dimension $2n$, the polarisation gives a K -hermitian form H on it of signature (n, n) , and the K -determinant

$$\bigwedge_K^{2n} V \subset \bigwedge_{\mathbb{Q}}^{2n} V = H^{2n}(A, \mathbb{Q})$$

is a plane of Hodge classes, the *Weil classes* [23]. Weil showed that on a general such A they are not products of divisor classes. In fact more is true: if the Hodge group of A is the whole special unitary group $\mathrm{SU}(V, H)$, which happens for a very general member of the family, then the Hodge ring of A is spanned by the powers of η together with this plane, and the Hodge conjecture for A in every degree is equivalent to the algebraicity of a single Weil class ω [16, 5].

These varieties sit in one family. Let D be the bounded symmetric domain of $\mathrm{SU}(n, n)$, of dimension n^2 , and let Γ be a torsion-free arithmetic subgroup of $\mathrm{SU}(V, H)(\mathbb{Q})$. Then $\mathrm{Sh} = \Gamma \backslash D$ is a smooth quasi-projective variety (Baily and Borel [4]), and over it there is a projective abelian scheme $\pi: \mathcal{A} \rightarrow \mathrm{Sh}$ whose fibres are the abelian varieties of Weil type with the given discriminant. Since Γ lies in the special unitary group, it acts trivially on the K -determinant. So the Weil classes form a *constant* local system of rank two inside $R^{2n}\pi_*\mathbb{Q}$, and ω is a global section of it.

Definition 1. The *algebraic locus* is the set $\Sigma \subseteq \text{Sh}$ of points t at which $\omega(t)$ is the class of an algebraic cycle with rational coefficients on A_t .

The goal is $\Sigma = \text{Sh}$. That statement, for every K , n and discriminant, is the Hodge conjecture for every abelian variety of Weil type. For $n = 2$ it is a theorem of Markman [12, 13], after earlier cases due to Schoen [20, 21]. For $n = 3$ and a split hermitian form it is also Markman's theorem, and Schoen and Koike treated particular families of sixfolds for $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-1})$ [20, 9]. Unrefereed preprints released in September 2026 announce the split eightfolds, and with a product argument of Schoen that would give every sixfold [5]. We use none of them. For $n \geq 4$, and for $n = 3$ in general, the question is open.

2. WHAT IS ALREADY KNOWN ABOUT THE LOCUS

The first three facts are proved in [5]. We include proofs of the first two, because they are short and the rest of the note uses the argument.

Lemma 2. Σ is a countable union of closed algebraic subsets of Sh .

Proof. Fix a relatively ample line bundle on $\mathcal{A}/\text{Sh}$. For each Hilbert polynomial P the relative Hilbert scheme $\text{Hilb}_P(\mathcal{A}/\text{Sh})$ is projective over Sh and has finitely many irreducible components. Take finitely many components $H_1, \dots, H_r$, possibly from different Hilbert schemes, and rational numbers $q_1, \dots, q_r$. Over the fibre product $H = H_1 \times_{\text{Sh}} \dots \times_{\text{Sh}} H_r$ we have the cohomology class $\sum_i q_i [Z_i]$ in the fibre $H^{2n}(A_t, \mathbb{Q})$. Classes of the fibres of a flat family of subschemes are locally constant. So on each connected component of H the class $\sum q_i [Z_i]$ either equals $\omega(t)$ everywhere or nowhere. The union of the components where it equals $\omega(t)$ maps properly to Sh , so its image is closed and algebraic. There are countably many choices of P 's, components and q_i 's, and every point of Σ lies in one of these images. $\square$

Lemma 3 (Dichotomy). *Either $\Sigma = \text{Sh}$, or Σ is meagre: it lies in a countable union of closed subsets with empty interior, and its complement is dense.*

Proof. Sh is irreducible. If one of the closed sets of Lemma 2 is all of Sh , then $\Sigma = \text{Sh}$. Otherwise each of them is a proper closed algebraic subset, which has empty interior, and Baire's theorem on the complex manifold Sh gives the rest. $\square$

Proposition 4 ([5]). Σ contains an explicit point s_0 at which A_{s_0} is isogenous to a product of abelian varieties with complex multiplication, and at which ω is a polynomial in divisor classes. Its orbit under the Hecke correspondences lies in Σ and is dense in Sh .

So Σ is dense, and the problem is entirely about which side of the dichotomy we are on. One stratum of Lemma 2 equal to all of Sh would settle it.

There is a catch, and it is the reason the problem is hard. Write Sh° for the set of points where the Hodge group of A_t is all of $\text{SU}(V, H)$. Its complement is a countable union of proper special subvarieties, so Sh° is the complement of a meagre set. Every point in the Hecke orbit of s_0 is a CM point, with a torus for its Hodge group, so it lies *outside* Sh° . To close the locus we need at least one point of $\Sigma \cap \text{Sh}^\circ$. For $K = \mathbb{Q}(\sqrt{-3})$ and a split form such points are known. The Prym varieties of étale cyclic triple covers of curves of genus $n + 1$ are of Weil type, Schoen proved that their Weil classes are algebraic [20], and a very general one has full Hodge group, by Looijenga's computation of the monodromy [11] and André's theorem. (This account follows [5].) For other fields and discriminants no such point is known when $n \geq 3$.

3. THE FIRST ROUTE: STRATA WITH BIG MONODROMY

Suppose that we had found one point $t \in \Sigma \cap \text{Sh}^\circ$, and let S be an irreducible stratum of Lemma 2 through it. The first thing to ask is what S can look like. The following lemma says that S cannot be a small, degenerate piece of the family.

Theorem 5. *Let $S \subseteq \text{Sh}$ be an irreducible closed subvariety of positive dimension that contains a point of Sh° . Then the connected algebraic monodromy group of the local system $R^1\pi_*\mathbb{Q}$ restricted to the smooth locus of S is the whole group $\text{SU}(V, H)$.*

Proof. Let S^{reg} be the smooth locus of S . It is connected, since S is irreducible. The Hodge group of a very general point of S contains the Hodge group of every point of S , so it contains $\text{SU}(V, H)$. On the other hand every Hodge group in the family lies in $\text{SU}(V, H)$ [5]. So the generic Hodge group on S is $\text{SU}(V, H)$.

By André's theorem [2], the connected algebraic monodromy group M of a polarisable variation of Hodge structure over a smooth connected base is a normal subgroup of the derived group of the generic Mumford–Tate group. Here that derived group is $\text{SU}(V, H)$, which is an outer form of SL_{2n} over $\mathbb{Q}$, and so it is almost simple. Its connected normal subgroups are 1 and itself.

If $M = 1$, the monodromy is finite. After a finite étale cover of S^{reg} the local system becomes trivial, and the period map lifts to a holomorphic map from that cover to D . Now D is a bounded domain, so the coordinate functions of the lift are bounded holomorphic functions. Pass to a resolution and a smooth compactification of the cover. By the Riemann extension theorem the bounded functions extend to the compactification, which is compact and connected, so they are constant. Then the period map is constant on S^{reg} . But the period map of the universal family over Sh is the identity of $\Gamma \backslash D$, so S is a point, which contradicts $\dim S > 0$. Hence $M = \text{SU}(V, H)$. $\square$

So a positive-dimensional stratum through a general point carries all the monodromy the family has. It is natural to hope that a cycle family with this much monodromy can be moved off S . That hope is exactly the variational Hodge conjecture for this family, and here is why the monodromy does not help on its own. The class ω is invariant under all of $\text{SU}(V, H)$, so the monodromy never moves ω , and it gives no constraint that would force the stratum to grow. Big monodromy rules out degenerate strata. It does not produce new cycles.

4. THE SECOND ROUTE: DEFORMING ONE CYCLE

The direct way to move a cycle is deformation theory. Let $Z \subset A_t$ be a local complete intersection of codimension n . Bloch [6] defined the *semiregularity map*

$$\sigma_Z: H^1(Z, N_Z) \longrightarrow H^{n+1}(A_t, \Omega^{n-1}),$$

and proved the following. If σ_Z is injective, then Z deforms along every deformation of A_t along which its class stays of Hodge type. Buchweitz and Flenner [7] extended the definition and the theorem to perfect complexes, and Pridham [18] gave a general proof.

Theorem 6. *Suppose that at some point $t \in \text{Sh}$ there is a local complete intersection $Z \subset A_t$ of codimension n whose class is $\text{cl}(Z) = \alpha + c\eta^n$ with α a nonzero Weil class and $c \in \mathbb{Q}$, and whose semiregularity map is injective. Then $\Sigma = \text{Sh}$.*

Proof. Along the universal family over a small neighbourhood U of t , the flat transport of $\text{cl}(Z)$ is $\alpha + c\eta^n$, and both terms are Hodge classes at every point, because the Weil plane and η are. So the class of Z stays of Hodge type along every direction of U . By Bloch's theorem, Z deforms to every order along every curve germ in U . Then the image of its Hilbert scheme component contains the formal neighbourhood of t . Since that image is a closed analytic subset of U , it

contains a neighbourhood of t . So one stratum of Lemma 2 has nonempty interior, and it is all of Sh by irreducibility. At every point s we then have an algebraic cycle of class $\alpha(s) + c\eta^n$, so $\alpha(s)$ is algebraic.

It remains to pass from α to ω . The field K acts on the Weil plane: an element $\tau \in K^\times$ acts on $\bigwedge_K^{2n} V$ by multiplication by τ^{2n} . A positive integer multiple $m\tau$ is a genuine endomorphism of A_s . Its pullback, divided by m^{2n} , is τ^* on H^{2n} , so τ^* carries algebraic classes to algebraic classes. Choose τ whose argument is not a rational multiple of π , for example $\tau = a + \sqrt{-d}$ with a a positive integer chosen so that $\tau/\bar{\tau}$ is not a root of unity. Then $\tau^{2n} \notin \mathbb{Q}$, and the classes α and $\tau^*\alpha$ span the Weil plane over $\mathbb{Q}$, so $\omega(s)$ is algebraic. $\square$

The same holds with Z replaced by a perfect complex E on A_t , with the Chern character in place of the cycle class, by the theorem of Buchweitz–Flenner and Pridham. This is the route Markman followed for sixfolds of split Weil type, using secant sheaves and an equivariant form of semiregularity [13].

5. WHERE THE ATTEMPT STOPS

Both routes need an input that we do not have.

The first route needs a point of Σ with full Hodge group. Outside the families where the conjecture is already known and Schoen’s Prym locus for $\mathbb{Q}(\sqrt{-3})$, every point at which ω is known to be algebraic is a CM point or lies on a proper special subvariety. These points are excluded from Sh° by definition. Theorem 5 tells us what a stratum through a point of Sh° looks like, but it gives no way to find one. Where one is known, Schoen’s Prym locus, it has dimension $3n$, while Sh has dimension n^2 . Nothing in the first route makes such a stratum grow.

The second route needs a semiregular object, and the obvious ones are not. For an object whose Chern character is *exactly* a Weil class, with no other component, the semiregularity map is never injective, at any member of any Weil family ([5], the theorem labelled `thm:p2false` in its source). The reason is Voisin’s. Such an object would deform along every K -linear deformation of A_t , including the non-algebraic complex tori of Weil type. A very general such torus has no analytic subvarieties of intermediate dimension, and its coherent sheaves have no Chern classes in intermediate degree [22]. So the deformation cannot exist, and the semiregularity map cannot be injective. With a polarisation term allowed, as in Theorem 6, the criterion survives, but it asks for an object with a precise and large amount of self-extension. For $n \geq 3$ and a general CM field no such object is known. A list of the natural constructions that provably fail is in [5].

Neither gap is a technicality. If $\Sigma = \text{Sh}$, then Σ meets Sh° , so the first gap is a consequence of the goal as well as a step towards it. And by [5] (the corollary labelled `cor:weilequiv`), closing the locus is equivalent to one case of the Lefschetz standard conjecture: the case of the total space of the family over a projective curve through s_0 and a very general point. These are the variational Hodge conjecture and the standard conjectures in disguise. They are not small residuals.

6. THE VARIATIONAL FORM OF THE PROBLEM

The question whether $\Sigma = \text{Sh}$ is one instance of the variational Hodge conjecture, and it helps to look at it that way. Let $f: X \rightarrow S$ be a smooth projective morphism onto a smooth connected variety, and let α be a global section of $R^{2p}f_*\mathbb{Q}$. By Deligne’s theorem of the fixed part [8], the global sections carry a Hodge structure that does not depend on the point, and the inclusion into each fibre is a morphism of Hodge structures. So if $\alpha(s)$ is a Hodge class at one point, it

is a Hodge class at every point. The variational Hodge conjecture asks for the corresponding statement about cycles: if $\alpha(s_0)$ is algebraic at one point s_0 , then $\alpha(s)$ is algebraic at every point s .

In our family, ω is a global section, it is a Hodge class everywhere, and by Proposition 4 it is algebraic at s_0 . The variational conjecture for $\pi: \mathcal{A} \rightarrow \text{Sh}$ and this one class says exactly that $\Sigma = \text{Sh}$. Three things are known about the conjecture in general, and none of them settles this case.

- (a) It holds when $p = 1$, by the theorem of Lefschetz on $(1, 1)$ classes. Here $p = n \geq 2$.
- (b) It follows from the Hodge conjecture for a smooth projective compactification $\bar{X}$ of the total space. By the global invariant cycle theorem [8] there is a class on $\bar{X}$ restricting to $\alpha(s)$, by semisimplicity of polarisable Hodge structures it can be chosen to be a Hodge class, and the restriction of a cycle representing it represents $\alpha(s)$ at every s . For our family $\bar{X}$ is a variety of dimension $2n + n^2$, or $2n + 1$ after restricting to a curve, and its Hodge conjecture is no easier than the one we want.
- (c) André's deformation principle [3] says that a flat section which is motivated at one point is motivated at every point, and André also showed that every Hodge class on an abelian variety is motivated. So $\omega(t)$ is already known to be motivated for every t . A motivated class is algebraic as soon as the Lefschetz standard conjecture holds for the varieties that enter its construction. This is the same reduction to the standard conjectures as in Section 5, reached from the other side.

The rest of this section contains three results. The first changes the question into a question about degrees. It is a variant, with Chow varieties and arbitrary Zariski-dense sets, of the bounded criterion of [5] (its theorems `thm:degreebound`, `thm:plequivalent` and `thm:escape`), which is stated for Hilbert data along Hecke orbits. The second shows that nothing built from divisors reaches ω at a member with full Hodge group. Its proof is the argument that [5] gives for Mumford's exceptional classes (its theorem `thm:lefschetzclosure`), carried over to the Weil family. The third is the analogue for the tautological cycles of curves with an automorphism of order three, and as far as I know it is new.

6.1. A criterion in terms of degrees. Measure the size of a cycle by its degree against the polarisation: for an effective cycle Z of dimension n on A_t , put $\deg_\eta Z = \int_{A_t} \text{cl}(Z) \cup \eta^n$. Every cycle with rational coefficients can be written as $\frac{1}{m}(Z^+ - Z^-)$ with $Z^\pm$ effective and m a positive integer. For $N \geq 1$ let Σ_N be the set of points t at which

$$\omega(t) = \frac{1}{m}(\text{cl}(Z^+) - \text{cl}(Z^-))$$

for some effective cycles $Z^\pm$ on A_t of dimension n with $\deg_\eta Z^\pm \leq N$ and some integer $1 \leq m \leq N$.

Proposition 7. *Each Σ_N is a closed algebraic subset of Sh , and $\Sigma = \bigcup_N \Sigma_N$. The following are equivalent.*

- (i) $\Sigma = \text{Sh}$.
- (ii) $\Sigma_N = \text{Sh}$ for some N .
- (iii) For some N , the set Σ_N contains a Zariski-dense subset of Sh .

Proof. Effective cycles of dimension n and bounded degree in the fibres of the projective family $\mathcal{A} \rightarrow \text{Sh}$ are parametrised by a relative Chow variety which is of finite type and proper over Sh [10, Ch. I]. Over a connected component of it, the class of the cycle is a flat section of $R^{2n}\pi_*\mathbb{Q}$, so the condition that $\frac{1}{m}(\text{cl } Z^+ - \text{cl } Z^-) = \omega$ holds on whole connected components of the fibre product of two such Chow varieties, or nowhere on them. Taking the union over $m \leq N$ of

the components where it holds and projecting to Sh gives Σ_N , which is therefore closed and algebraic. Every point of Σ lies in some Σ_N .

(i) implies (ii): Sh is the countable union of the closed sets Σ_N , so by Baire's theorem one of them has nonempty interior, and since Sh is irreducible that one is all of Sh . (ii) implies (iii) trivially. (iii) implies (i): a Zariski-closed set that contains a Zariski-dense set is everything. $\square$

The proposition is elementary, but it changes the shape of the question, as [5] already observed in its own setting. Proposition 4 already gives algebraic representatives of ω on a dense set. What is missing is a *uniform bound* on their degrees and denominators along some Zariski-dense subset. Transport along isogenies does not give one, and it is worth seeing why.

Remark 8. Let $\phi: A_t \rightarrow A_{t'}$ be an isogeny of degree d that commutes with K and satisfies $\phi^*\eta' = e\eta$ for an integer $e \geq 1$, as the Hecke correspondences do. The two polarisations have the same type, so comparing Euler characteristics gives $e^{2n}\chi(\eta) = \chi(\phi^*\eta') = d\chi(\eta') = d\chi(\eta)$, that is $d = e^{2n}$. For an effective cycle Z of dimension n the projection formula gives

$$\deg_{\eta'} \phi_* Z = \int_{A_t} \text{cl}(Z) \cup \phi^*\eta'^n = e^n \deg_{\eta} Z.$$

An abelian variety has only finitely many isogenies of bounded degree, so along any infinite subset of the Hecke orbit of s_0 the multipliers e are unbounded, and so are the degrees of the transported cycles. Proposition 7 asks for *other* cycles at those points, of bounded degree. The Hecke construction gives no hint where to find them.

6.2. A barrier at members with full Hodge group. Fix $t \in \text{Sh}^\circ$, write $A = A_t$, and let $U = U(V, H)$ be the unitary group of the hermitian form, as an algebraic group over $\mathbb{Q}$. It acts on $V = H^1(A, \mathbb{Q})$, and so on $H^*(A, \mathbb{Q}) = \bigwedge^* V$. Let $\mathcal{C}$ be the category of abelian varieties isogenous to a power of A , with homomorphisms as morphisms. Choosing an isogeny $B \rightarrow A^m$ lets U act on $H^1(B, \mathbb{Q}) \cong V^m$ diagonally. Two choices differ by an invertible element of $\text{End}^0(A^m)$, and we shall see in a moment that these commute with U , so the action does not depend on the choice.

Theorem 9. *Let $n \geq 2$ and $t \in \text{Sh}^\circ$. Let $(\mathcal{L}(B))_{B \in \mathcal{C}}$ be the smallest family of subspaces $\mathcal{L}(B) \subseteq H^*(B, \mathbb{Q})$ such that each $\mathcal{L}(B)$ is a subalgebra containing the classes of all divisors on B , and $h^*\mathcal{L}(B') \subseteq \mathcal{L}(B)$ and $h_*\mathcal{L}(B) \subseteq \mathcal{L}(B')$ for every homomorphism $h: B \rightarrow B'$ in $\mathcal{C}$. Then every class in every $\mathcal{L}(B)$ is invariant under U . In particular no class in $\mathcal{L}(A)$ has the form $\alpha + c\eta^n$ with α a nonzero Weil class.*

Proof. Step 1: the endomorphisms. Over $\mathbb{C}$ the space V splits as $V_\sigma \oplus V_{\bar{\sigma}}$, the two eigenspaces of K , each of dimension $2n$. The polarisation form E is invariant under the norm-one elements of K , which forces V_σ and $V_{\bar{\sigma}}$ to be isotropic and E to pair them perfectly. So, as representations of $\text{SU}(V, H)_\mathbb{C} \cong \text{SL}_{2n}$, they are the standard representation and its dual. For $2n \geq 3$ these are irreducible and not isomorphic, so the commutant of $\text{SU}(V, H)$ in $\text{End}_\mathbb{Q}(V)$ has dimension two. Since the Hodge group of A is $\text{SU}(V, H)$, this commutant is $\text{End}^0(A)$, and so $\text{End}^0(A) = K$ and $\text{End}^0(A^m) = M_m(K)$. A homomorphism $h: A^m \rightarrow A^k$ therefore acts on H^1 by a matrix with entries in K , and since U acts K -linearly on V , the map $h^*: V^k \rightarrow V^m$ commutes with the diagonal action of U . The same holds in $\mathcal{C}$ after composing with isogenies.

Step 2: the centre. The centre of U is the torus $U(1)_K$ of norm-one elements of K , and $U = \text{SU}(V, H) \cdot U(1)_K$. An element λ of the centre acts on V_σ by $\sigma(\lambda)$ and on $V_{\bar{\sigma}}$ by $\sigma(\lambda)^{-1}$, so on the summand $\bigwedge^a V_\sigma^{\oplus m} \otimes \bigwedge^b V_{\bar{\sigma}}^{\oplus m}$ of $H^{a+b}(A^m, \mathbb{C})$ it acts by $\sigma(\lambda)^{a-b}$. Hence a class is U -invariant exactly when it is $\text{SU}(V, H)$ -invariant and lies in the sum of the summands with $a = b$.

Step 3: divisors. A divisor class on A^m is a Hodge class, and the Hodge group of A^m is $\mathrm{SU}(V, H)$ acting diagonally, so the class is $\mathrm{SU}(V, H)$ -invariant. Its components with $a \neq b$ lie in $\bigwedge^2(V_\sigma \otimes \mathbb{C}^m)$ or its conjugate. As a representation of SL_{2n} this is a sum of copies of $\bigwedge^2 V_\sigma$ and $\mathrm{Sym}^2 V_\sigma$, which have no invariants when $2n \geq 3$. So those components vanish, and by Step 2 the class is U -invariant.

Step 4: the operations. Cup product and h^* commute with U , by Step 1. Every element of U preserves E , so it has determinant one on V and acts trivially on the top cohomology of each B . So U preserves the Poincaré pairing, and h_* , being the adjoint of h^* for that pairing, commutes with U as well. It follows that the family of U -invariant subalgebras $H^*(B, \mathbb{Q})^U$ has all the properties in the statement, and by minimality $\mathcal{L}(B) \subseteq H^*(B, \mathbb{Q})^U$.

Step 5: Weil classes. The Weil plane, tensored with $\mathbb{C}$, is $\bigwedge^{2n} V_\sigma \oplus \bigwedge^{2n} V_{\bar{\sigma}}$, the summands with $(a, b) = (2n, 0)$ and $(0, 2n)$. The centre acts on them by $\sigma(\lambda)^{\pm 2n}$. Taking $\lambda = \tau/\bar{\tau}$ with τ as in the proof of Theorem 6, this is not 1, so the Weil plane has no nonzero U -invariant vector. Since η^n is U -invariant, a U -invariant class $\alpha + c\eta^n$ has α U -invariant, hence $\alpha = 0$. $\square$

The theorem reaches further than it looks, because Chern characters behave well under the same operations.

Corollary 10. *Let $t \in \mathrm{Sh}^\circ$, and let F be an object of the derived category of A_t obtained from line bundles on varieties in $\mathcal{C}$ by finitely many shifts, direct sums, cones, derived tensor products, and derived pullbacks and pushforwards along homomorphisms. Then $\mathrm{ch}(F)$ is U -invariant, and F cannot satisfy the hypothesis of Theorem 6. This applies in particular to Fourier–Mukai transforms whose kernel is the Mumford line bundle $m^*L \otimes \mathrm{pr}_1^*L^{-1} \otimes \mathrm{pr}_2^*L^{-1}$ on $A_t \times A_t$, and to pushforwards of line bundles along isogenies, which by a theorem of Mukai include all simple semi-homogeneous vector bundles [15].*

Proof. The Chern character is additive on exact triangles and multiplicative on derived tensor products, it commutes with pullback, and $\mathrm{ch}(L) = \exp c_1(L)$ for a line bundle. For pushforward along a homomorphism h , the Grothendieck–Riemann–Roch theorem gives $\mathrm{ch}(Rh_*F) = h_*\mathrm{ch}(F)$, because the Todd class of an abelian variety is 1. So $\mathrm{ch}(F)$ lies in $\mathcal{L}(A_t)$, and Theorem 9 applies. $\square$

So at a member with full Hodge group, nothing built from line bundles and homomorphisms has the right Chern character. This explains a feature of every successful construction so far. At s_0 the endomorphism algebra is large, the corresponding group is smaller than U , and ω is a polynomial in divisor classes. Markman’s secant sheaves also start on special members, built from an abelian n -fold and its dual, and reach the general member only by deformation [12, 13]. The theorem shows that this detour is forced: a proof has to start where the endomorphisms are large and then deform, or it has to find at a general member a cycle that does not come from divisors at all.

6.3. Curves with an automorphism of order three. The oldest source of cycles that do not come from divisors is curves. For $K = \mathbb{Q}(\sqrt{-3})$ there are two natural families of curves whose Jacobians contain abelian varieties of Weil type, and one might hope that cycles built from the curve reach ω . The next proposition shows that the obvious cycles do not.

Let C be a smooth projective curve with an automorphism ρ of order three, and let $V \subseteq H^1(C, \mathbb{Q})$ be the kernel of $1 + \rho^* + \rho^{*2}$, the part on which ρ has no fixed vectors. Then K acts on V through ρ . Let $P \subseteq J(C)$ be the image of $1 - \rho$, so that $H^1(P, \mathbb{Q})$ is identified with V , and assume that P , with the polarisation induced from $J(C)$, is of Weil type for K . Let $U = U(V, H)$ act on $H^1(C, \mathbb{Q}) = V \oplus H^1(C, \mathbb{Q})^\rho$, trivially on the second summand, and trivially on H^0 and H^2 .

Proposition 11. *Let $R^*(C^m) \subseteq H^*(C^m, \mathbb{Q})$ be the subring generated by the classes of the diagonals $\{x_i = x_j\}$, the graphs $\{x_j = \rho^k(x_i)\}$ for $k = 1, 2$, and the pullbacks of the point class from each factor. Let $f: C^m \rightarrow P$ be a morphism of the form $h \circ (\text{AJ} \times \cdots \times \text{AJ})$, where AJ is an Abel–Jacobi map $C \rightarrow J(C)$ and $h(y_1, \dots, y_m) = (1 - \rho) \sum_i \psi_i(y_i)$ with $\psi_i \in \mathbb{Z}[\rho]$. Then every class in $f_* R^*(C^m)$ is U -invariant, and in particular none has the form $\alpha + c\eta^n$ with α a nonzero Weil class.*

Proof. The summand V is orthogonal to $H^1(C, \mathbb{Q})^\rho$ under cup product, because the pairing is ρ -invariant and ρ acts on the two summands with no common eigenvalue. The group U preserves the pairing on V , which is the polarisation form of P up to a constant. So the action of U on $H^1(C, \mathbb{Q})$ is symplectic, it extends to an action on $H^*(C^m, \mathbb{Q})$ by algebra automorphisms, and it fixes the top class.

The class of the graph of an endomorphism T of H^1 , in the $H^1 \otimes H^1$ part of $H^2(C \times C)$, is the tensor corresponding to T under Poincaré duality. It is invariant under every symplectic g that commutes with T . The diagonal is the case $T = 1$, and the graphs of ρ^k are the cases $T = \rho^{\pm k}$, which commute with U because U is K -linear. The other components of these classes, and the point classes, lie in $H^0 \otimes H^2$ and $H^2 \otimes H^0$, where U acts trivially. So $R^*(C^m)$ is U -invariant.

The pullback $f^*: H^1(P) = V \rightarrow H^1(C)^{\oplus m}$ is given in each coordinate by an element of $\mathbb{Z}[\rho]$, that is, by multiplication by an element of K . So it lands in $V^{\oplus m}$ and commutes with U , which acts K -linearly. (This is where the special form of h is used: a general K -linear map of V does not commute with U .) Hence f^* commutes with U on all of $H^*(P)$, and f_* , its adjoint for the Poincaré pairings, does too. The last assertion follows as in Step 5 of the proof of Theorem 9. $\square$

Two families fit the hypotheses. In the first, C is the cyclic triple cover $y^3 = \prod_{i=1}^{n+1} (x - a_i) \prod_{j=1}^{n+1} (x - b_j)^2$ of the projective line. By Riemann–Hurwitz it has genus $2n$, and by the Chevalley–Weil formula the two nontrivial eigenspaces of ρ on holomorphic differentials both have dimension n , so $P = J(C)$ is of Weil type. These curves depend on $2n - 1$ parameters, which is less than n^2 for $n \geq 2$, so their Jacobians fill only a proper subvariety of Sh . In the second, $C \rightarrow C'$ is an étale cyclic triple cover of a curve of genus $n + 1$. Then C has genus $3n + 1$, the two nontrivial eigenspaces both have dimension n , and P is the Prym variety, of dimension $2n$. These depend on $3n$ parameters, and for $n \leq 3$ that is at least n^2 . Here much more is known than the proposition might suggest. Schoen proved that the Weil classes of these Prym varieties are algebraic [20], so for $n \leq 3$ the Pryms give whole families of Weil type in which the conjecture holds, and for $n \geq 4$ they give a locus of dimension $3n$ inside Sh on which ω is algebraic and whose very general point has full Hodge group. The proposition says something about how this can happen. Schoen’s cycles represent a nonzero Weil class, so they cannot be the image of a tautological class under a map of the form in the proposition. Whatever they are built from, it is a cycle on a power of C , or a map, outside the tautological world. That is exactly the kind of source that Theorem 9 says a proof needs, and it is the most concrete lead this note can offer (see the next subsection).

6.4. What the new results leave. Put together, the three results narrow the search. At a member with full Hodge group, ω cannot be reached by anything built from divisors and homomorphisms (Theorem 9 and Corollary 10), nor by the tautological cycles of a curve with an automorphism of order three (Proposition 11). At special members it can be reached, and Hecke transport spreads those representatives over a dense set, but with degrees that the transport itself drives to infinity, and Proposition 7 says that bounded degree on a Zariski-dense set is exactly what is needed. Any proof therefore has to supply one of three things: a uniform bound on degrees along a Zariski-dense set of special members; a deformation argument that carries a cycle from a special member to the general one, which is the semiregularity route of Theorem 6;

or a source of cycles at a general member that is neither divisorial nor tautological. The third exists for $\mathbb{Q}(\sqrt{-3})$: Schoen's cycles on Prym varieties. Their locus has dimension $3n$, so for $n \geq 4$ they reach only a thin part of Sh. The obvious question is whether one of Schoen's cycles is semiregular. The construction makes the question concrete. On the symmetric power $C'^{(2n)}$ of the base curve, the canonical system $|K_{C'}| \cong \mathbb{P}^n$ is the fibre of the Abel–Jacobi map over the canonical class. The triple cover defines an étale cover of $C'^{(2n)}$ of degree three, and over $|K_{C'}|$, which is simply connected, it splits into three copies of $\mathbb{P}^n$. The differences of the copies, weighted by the characters of order three, are Schoen's cycles [20, Theorem 2.0 with $r = 0$], and Patel and Zhang reach the same copies by base change along an isogeny onto the Jacobian [17, Theorem 1.1]. On the Prym the class of one copy is a Weil class plus a multiple of η^n . When that multiple is nonzero, this is exactly the shape that Theorem 6 allows and that the obstruction from Weil tori does not exclude. If one of them were semiregular, Theorem 6 would give $\Sigma = \text{Sh}$ for the family of that dimension. The construction also suggests why this may fail. The Abel–Jacobi curve $C \subset J(C)$ comes from the same map, and its class $\theta^{g-1}/(g-1)!$ is algebraic on every principally polarized abelian variety; yet by the criterion of Matsusaka and Ran a curve with this class exists only on Jacobians and products of Jacobians [14, 19]. For $g \geq 4$ these fill a proper subvariety of the moduli space, so by Bloch's theorem the Abel–Jacobi curve is not semiregular. Schoen's cycles are the analogue one step up, and nothing in their construction moves them off the Prym locus. This is an analogy, not a proof, and I have not been able to decide the question. Outside $\mathbb{Q}(\sqrt{-3})$, none of the three is available for $n \geq 3$ in general.

7. WHAT THIS NOTE ESTABLISHES

Proved here, or proved in the cited work and used as theorems:

- (1) The algebraic locus is a countable union of closed algebraic subsets, it is dense, and it is either everything or meagre.
- (2) Any positive-dimensional stratum of it through a member with full Hodge group has full monodromy $\text{SU}(V, H)$.
- (3) A single semiregular cycle or complex, at any member, with class a nonzero Weil class plus a multiple of η^n , would close the locus, and with it the Hodge conjecture for every abelian variety of Weil type in that family.
- (4) The problem is the variational Hodge conjecture for one class, and it holds if and only if ω has representatives of bounded degree and denominator on a Zariski-dense set of members. Hecke transport does not give such a bound.
- (5) At a member with full Hodge group, no class built from divisors by cup products, pullbacks and pushforwards along homomorphisms, and no Chern character of an object built from line bundles in the same way, has a nonzero Weil component.
- (6) On a curve with an automorphism of order three, the cycles built from diagonals, graphs of the automorphism and the group law push forward to classes with no Weil component.

Not proved: that such a cycle or complex exists when $n \geq 3$, outside the families where the conjecture is already known; that Schoen's cycles on Prym varieties are semiregular; or, for fields other than $\mathbb{Q}(\sqrt{-3})$ with a split form, that Σ contains a single member with full Hodge group. So the Hodge conjecture for abelian varieties of Weil type remains open in dimension eight and above, and in dimension six outside the known families. The Hodge conjecture itself remains open.

Machine checks. No proof in this note depends on a computer. The finite steps of Theorem 9 and of the discussion after Proposition 11 have nevertheless been repeated by machine, and the files are in the author's repository [1].

In Lean 4, checked by its kernel with nothing beyond the core library, we verified three things. For every squarefree $d \leq 200$ the element $\tau = 2 + \sqrt{-d}$ satisfies $\tau^{12} \neq \bar{\tau}^{12}$. Since every root of unity in an imaginary quadratic field has twelfth power 1, this shows that $\tau/\bar{\tau}$ is not a root of unity, as Step 5 requires. (For $d = 4$ and $d = 12$, which are not squarefree, this choice of τ fails.) The weights of Steps 2 and 5 were checked as well, and so were the genus, eigenspace and parameter counts of the two families of curves.

In Julia, and in a Python version of the same checks, both in exact arithmetic, we computed two things. For $2n \leq 6$ and $m \leq 2$ the endomorphisms of $H^1(A^m)$ commuting with $\mathrm{SU}(V, H)$ span $2m^2$ dimensions, as Step 1 says. For $2n \leq 6$ and $m \leq 3$ the invariants in degree two on A^m form m^2 classes of type $(1, 1)$, with none of type $(2, 0)$ or $(0, 2)$, as Step 3 says. For $2n = 2$ both computations come out differently, as they should, since the hypothesis $2n \geq 3$ is then false.

None of this touches the geometry of the note, and none of it bears on the open questions of Section 5.

REFERENCES

- [1] D. Bhattacharjee, *Hodge-Conjecture-Full*, repository of the Lean and Julia checks, 2026, <https://github.com/creeelie/Hodge-Conjecture-Full>, directory `verification`.
- [2] Y. André, *Mumford–Tate groups of mixed Hodge structures and the theorem of the fixed part*, *Compositio Math.* **82** (1992), 1–24.
- [3] Y. André, *Pour une théorie inconditionnelle des motifs*, *Publ. Math. IHÉS* **83** (1996), 5–49.
- [4] W. L. Baily, Jr. and A. Borel, *Compactification of arithmetic quotients of bounded symmetric domains*, *Ann. of Math.* (2) **84** (1966), 442–528.
- [5] D. Bhattacharjee, *Algebraic Loci of Weil Classes on Abelian Varieties*, preprint, Zenodo (2026), version 5.1.0, <https://doi.org/10.5281/zenodo.23227940>.
- [6] S. Bloch, *Semi-regularity and de Rham cohomology*, *Invent. Math.* **17** (1972), 51–66.
- [7] R.-O. Buchweitz and H. Flenner, *A semiregularity map for modules and applications to deformations*, *Compositio Math.* **137** (2003), 135–210.
- [8] P. Deligne, *Théorie de Hodge II*, *Publ. Math. IHÉS* **40** (1971), 5–57.
- [9] K. Koike, *Algebraicity of some Weil Hodge classes*, *Canad. Math. Bull.* **47** (2004), 566–572.
- [10] J. Kollár, *Rational Curves on Algebraic Varieties*, *Ergebnisse der Mathematik und ihrer Grenzgebiete* (3) **32**, Springer, Berlin, 1996.
- [11] E. Looijenga, *Prym representations of mapping class groups*, *Geom. Dedicata* **64** (1997), 69–83.
- [12] E. Markman, *Cycles on abelian $2n$ -folds of Weil type from secant sheaves on abelian n -folds*, preprint, arXiv:2502.03415 (2025).
- [13] E. Markman, *Secant sheaves and Weil classes on abelian varieties*, preprint, arXiv:2509.23403 (2025).
- [14] T. Matsusaka, *On a characterization of a Jacobian variety*, *Mem. Coll. Sci. Univ. Kyoto Ser. A Math.* **32** (1959), 1–19.
- [15] S. Mukai, *Semi-homogeneous vector bundles on an abelian variety*, *J. Math. Kyoto Univ.* **18** (1978), 239–272.
- [16] B. Moonen and Yu. G. Zarhin, *Hodge classes on abelian varieties of low dimension*, *Math. Ann.* **315** (1999), 711–733.
- [17] D. Patel and Y. Zhang, *Algebraicity of Hodge classes on some generalized Prym varieties*, preprint, arXiv:2506.13729, 2025.
- [18] J. P. Pridham, *Semiregularity as a consequence of Goodwillie’s theorem*, *Forum Math. Sigma* **12** (2024), e126.
- [19] Z. Ran, *On subvarieties of abelian varieties*, *Invent. Math.* **62** (1981), 459–479.
- [20] C. Schoen, *Hodge classes on self-products of a variety with an automorphism*, *Compositio Math.* **65** (1988), 3–32.
- [21] C. Schoen, *Addendum to: Hodge classes on self-products of a variety with an automorphism*, *Compositio Math.* **114** (1998), 329–336.
- [22] C. Voisin, *A counterexample to the Hodge conjecture extended to Kähler varieties*, *Int. Math. Res. Not.* **2002**, no. 20, 1057–1075.
- [23] A. Weil, *Abelian varieties and the Hodge ring*, in: *Œuvres Scientifiques*, Vol. III, Springer, New York, 1979, 421–429.